



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz 2 – FALL 2025

Course Title:	Digital System Design	Course Code:	CPE 344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22	Section:	A
Date:	13 th October, 2025		Maximum Marks:	10	
Student's Name:			Registration Number:		

Question 1:

[CLO1-PLO3-C5]

[10 Marks]

Design a system with an 8-bit input I that can be stored in 8-bit registers A, B, and/or C when input La, Lb, and/or Lc is 1, respectively. So if inputs La and Lb are 1, then registers A and B will be loaded with input I, but register C will keep its current value. Furthermore, if input R is 1, then the register values swap such that A=B, B=C, and C=A. Input R has priority over the L inputs. The system has one clock input also.

